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Context Aware Process Mining in Logistics

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Processes in manufacturing and logistics are characterized by a high frequency of changes and fluctuations, caused by the high number of participants in logistic processes and the variety of goods handled and services offered. This dynamic behavior particularly requires well documented processes, but at the same time it also complicates process documentation. Manually documented processes can, e.g., miss details of alternative process branches, and the continuing change in logistic processes renders the documentation quickly outdated. A possible solution approach is to automate the documentation of processes. This automated documentation, if based on transaction and master data from the IT systems connected to the logistic process, is called Process Mining. Being a subfield of Data Mining, Process Mining extracts sequences of activities from event logs in databases. Logistics has the opportunity to greatly benefit from the application of Process Mining, because the identification and tracking of goods in the supply chain involves many IT systems. However, the IT landscape in logistics is heterogeneous, because the data are scattered among different specialized systems for various purposes (e.g., warehousing, transportation planning, and billing) of different companies. Due to this lack of standardization, the data cannot simply be analyzed by a predefined routine. Therefore, additional information beside the stored data should be taken into consideration. Machine Learning offers the possibility to classify single items in large data sets and to categorize these items with regard to the context they are in. Adding context awareness to unstructured event data in logistics has the potential to improve the results of Process Mining. Our research investigates how to apply context awareness based on Machine Learning in Process Mining for logistic processes and demonstrates its performance in a logistic scenario.

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Keywords: Logistics, Process Mining, Context Awareness**1. Introduction**

Logistic and production systems are characterized by a high degree of complexity, dynamics, and uncertainty. The large numbers of participants in logistics processes as well as the variety of goods and service handled cause this complexity. At the same time, technological advances in combination with the globalization of activities in manufacturing lead to frequent innovation and shorter life cycles of products [1]. Therefore, logistic systems are required to adapt quickly to changes so that companies are able to gain competitive advantages [2], [3]. Enabling this reactivity is challenging due to the high level of dynamics and uncertainty in most logistic systems. Being aware of the actual logistic processes and deviations from the planned processes in a dynamic environment is essential for companies

in order to gain additional flexibility [2]. As a result, the documentation of processes is likely to become obsolete and lead to a mismatch between the real process and the documented process.

Process Mining is a potential tool to solve the challenges of keeping the process documentation up to date. It is the technique to work on business processes by discovering, monitoring, and improving actual processes by extracting knowledge from event logs [4]. The knowledge from Process Mining can be used to optimize the current processes which leads to higher competitiveness, e.g., by cost reduction, management time reduction, and the elimination of errors [4]. Process Mining has been successfully deployed to gain insight knowledge of business processes in a variety of application domains such as healthcare, education, software development,

and also logistics and production [5] [6]. Nevertheless, most studies examine data from simulated scenarios or for a single case. This paper deploys the Process Mining technique to analyze a larger number of heterogeneous logistic processes provided in datasets in order to investigate to what extent Process Mining can be automated.

Previous studies of Process Mining focus on extracting event logs to discover processes, check conformance, and enhance the processes [7] [6] [5]. However, Process Mining can go beyond analyzing historic data. There is important context data which is relevant in a logistic process. In order to further develop the Process Mining technique, context awareness is added to discover specific contexts based on the particular task. The aim of this paper is to investigate to what extent context information can support Process Mining in heterogeneous processes in manufacturing and logistics. In the subsequent section, we briefly review the key ideas of Process Mining, context awareness, and clustering. Section 3 proposes and tests an approach to improve Process Mining by adopting clustering techniques in the data pre-processing to manage event logs of high volume and with high variability in the records [8].

2. Related Work

2.1. Process Mining

Business process modelling is a visual way of presenting operations in organizations, e.g., entities, activities, and their relations [9], [10]. It has been widespread used within organizations. A number of research activities is devoted to process modeling techniques and tools [11]. Process Mining is the link between data mining, business process modeling, and analysis [12]. Software vendors more and more start to integrate Process Mining functionality to their tools [12]. Process Mining has furthermore been recognized as a potential tool for analyzing and improving processes also in logistics and manufacturing [5]. It exploits the knowledge from event logs which are generated by information systems. Making use of this knowledge is beneficial for process improvement [4]. Most Process Mining research is applied in the areas of healthcare, software development process, and online education [13].

In logistics and manufacturing, business process modelling plays a vital role [4], [11]. As a consequence of the complexity and rapid changes of the logistics processes, existing process models quickly become obsolete and can no longer serve their initial purpose. Process Mining was already proposed to be integrated into a wider Process Maintenance Framework to deal with heterogeneous logistic processes and automated process model creation effectively [5]. Process Mining in manufacturing and logistics has been tested already with simulated logistic data. It has been proven to be able to mine the actual logistic process, yet it needs improvements for a regular practice such as how to manage heterogeneous data sources [5]. However, a comprehensive understanding and procedure of Process Mining in logistic and manufacturing is still missing.

2.2. Context-Awareness in Logistics

A context-aware system is a system that is able to provide suitable service or information with regard to a user's task [14], [15]. There are many authors who define the meaning of context differently. In this paper we follow the definition of Dey (1999) as "Context is characteristics which describe the situation of an entity. An entity refers to a person, place, or object which is related to the interaction between user and application" [16].

The logistics and manufacturing domains are ideal candidates to include context awareness features, because industrial activities happen in very heterogeneous environments with a multitude of information available beside the actual observed process. Context data can be time, location, and frequency of events as well as related communication, tools, devices, or operators.

The Process Mining technique utilizes event logs as the primary input data. These events log represent sequentially recorded events. Each event refers to a single activity (a step in some process) and it is associated with a particular case (a process instance) [12]. All Process Mining techniques use these activities and cases for analysis. However, event logs may store additional data such as timestamps, size of order, related persons, etc. [12] Furthermore, additional context information can be extracted from event logs when moving away from investigating single log entries and proceeding to merging information on a higher level, e.g. grouping similar events and counting their frequency of occurrence. In our paper, we include the frequency of a process and its overall cycle time as the relevant context information as part of a logistics and manufacturing process. In Section 3 we will illustrate the influence of these pieces of context information on Process Mining in logistics and manufacturing.

2.3. Clustering Analysis

The purpose of data clustering is to group a set of objects based on their similarity of attributes and their dissimilarity to other groups [17]. In Process Mining, clustering analysis can be applied in the data pre-processing step to handle large amounts of data [18]. Trace clustering in Process Mining based on the generic edit distance has been deployed in research on Process Mining in the healthcare domain. The results illustrate that clustering techniques can be effective to handle spaghetti-like processes in the healthcare domain [18].

In the logistics and manufacturing domain, event logs are complex, heterogeneous, and may contain outliers. Furthermore, the sequence of activities or events is sensitive. Generally, additional information other than the events themselves can be retrieved from event logs, i.e. the number of events per case, unique events, unique paths, and the similarity among cases. The clustering method we apply in this paper is k-medoids, which is considered as a suitable method for clustering processes in logistics and manufacturing.

The k-medoids clustering method is a variant of the popular k-means algorithm, and they both are classified as partitioning clustering algorithms [19], [20]. In contrast to k-means, k-medoids is able to work with nominal attributes, because it uses

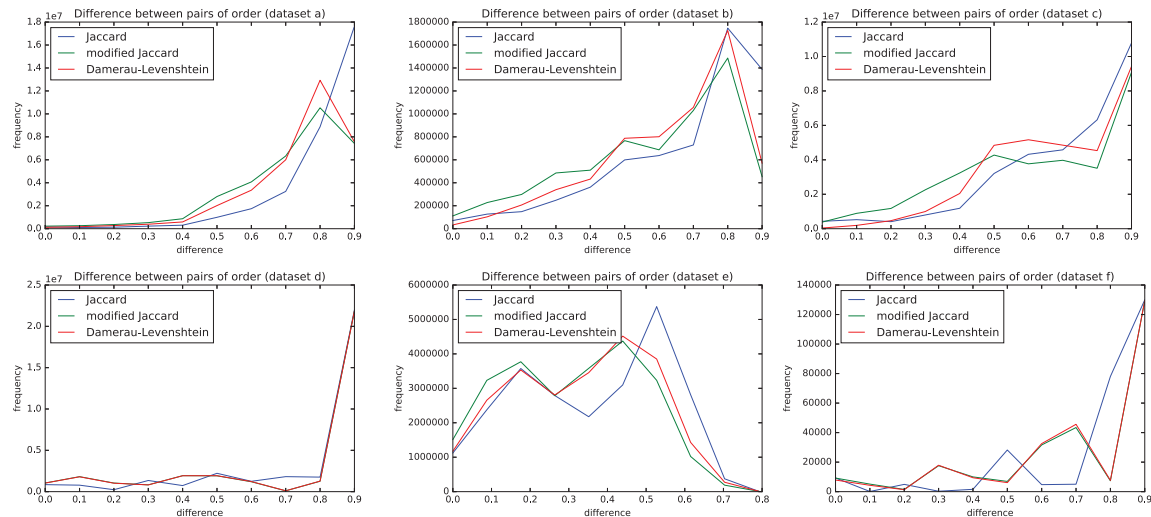


Fig. 1 Frequency distributions of the differences between each pair of processes as calculated with the three different distance measures. High values on the right side of a plot indicate rather different processes, while high values on the left side can be found if the processes within a dataset are more likely to be similar. All three different measures tested give similar results.

actual items from the dataset as centroids (instead of computing new cluster centroids as k-means does) and it is less sensitive to outliers [20]. K-medoids is an iterative process of clustering, and it works as follows:

- (1) First, the k-medoids algorithm selects k representative objects as cluster centers (medoids), e.g. via random selection. Then each object is assigned to the cluster with the closest medoid, using the chosen distance measure (e.g., Jaccard distance, Levenshtein distance, or Damerau-Levenshtein distance).
- (2) Subsequently, new centroids are computed by finding the object with the lowest summed distance to all other cluster members.
- (3) Each object is again assigned to the closest centroid.
- (4) Repeat the steps (2) and (3) until a stop criterion is satisfied (usually when no new centroids in (2) are found).

The k-medoids algorithm is deployed in our paper to cluster the processes, because of two important features: First, it is able to handle nominal data, and second it works with centroids for each cluster, which represent the center of a cluster. The first feature, the handling of nominal data, is required because processes are represented by a vector of activities, and each vector is a unique representation based on the sequence of nominal items. The second feature, the centroids, are necessary because the key idea of our approach is to identify main process, which represent cluster centers, whereas the other cluster members are considered as variants of the main process. As similarity measures are the basis of clustering, we also evaluate a selection of measures in our paper to identify the most suitable measure for our purpose. Again, we have to stick to measures that are able to handle data in the shape of vectors of varying length containing nominal items.

3. Methodology

3.1. Approach

The aim of our work is to improve the results of Process Mining in heterogeneous process environments, such as manufacturing and logistics. In particular, we are looking at scenarios where we can observe a certain number of ‘main’ processes in combination with a high amount of processes that are variations of the main processes. In a first step, we need to define the similarity (or the difference) of two processes to be able to judge if these processes are variations of the same type of main process or if the observed processes belong to two different groups. Therefore, we briefly revisit distance measures that have been applied to quantify the difference of traces in Process Mining, and we evaluate three different distance measures in order to identify the most suitable measure for our purpose. Consequently, we split the complete set of traces into clusters to separate the main processes from each other and to group them together with their respective variants. The clustering if the process data is not the desired result, it is just a vehicle to identify meaningful cluster centers, which in turn are identified as main processes as part of a process documentation effort. Usage scenarios in industry are, e.g., the support of process standardization efforts or actions to prioritize processes based on their importance.

We modify an existing clustering algorithm in a way that it incorporates contextual information about the frequency of occurrence of processes and about consistency in terms of cycle time. Finally, we use measures for cluster quality to assess whether the inclusion of contextual information can improve the identification of related processes.

We have selected six datasets provided by companies from the manufacturing and logistics services area. Each dataset

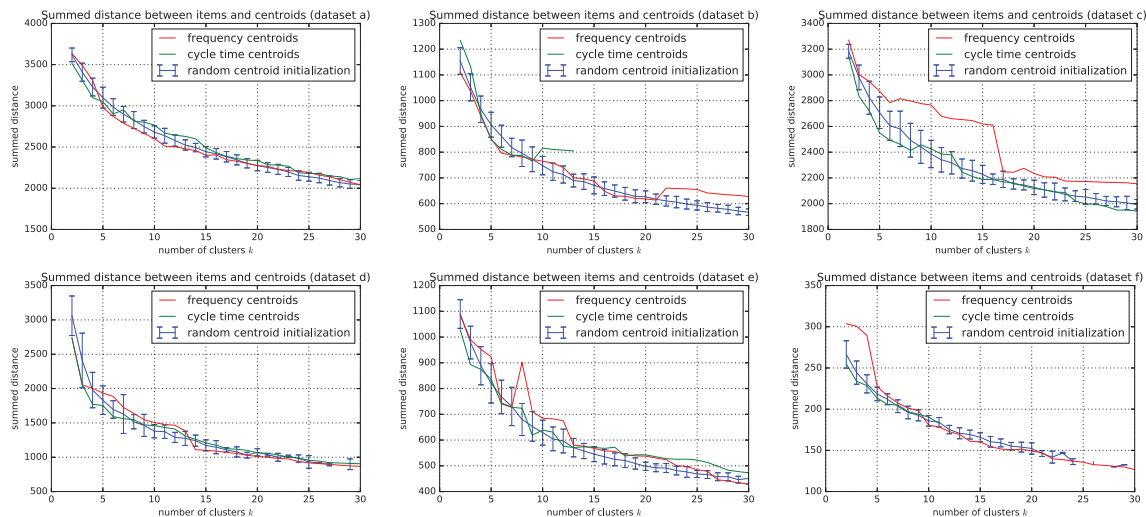


Fig. 2 Evaluation of the clustering quality for the two context based approaches (frequency and cycle time) in comparison to the random initialization of centroids in the k-medoids algorithm. The error bars of the 'random initialization' plot indicate the standard deviation over the 100 clustering runs.

contains a list of operations, which have occurred in a predefined period. The operations records contain an operation ID, an order ID, the work system (e.g., a machine on the shop floor or a storage unit), and two timestamps indicating the start and the end of the operation. An order is composed of at least one (usually more) consecutive operations. In the terms of Process Mining as introduced by van der Aalst [4], one dataset is an event log and each order (or process) is a trace, composed of several events.

3.2. Modeling of the Process Information

The processes we observe in the data are lists of consecutive operations in a manufacturing environment. We assume that two operations are identical if they are carried out on the same work system (machine). Therefore, we can express each order as a vector of machines of varying length. If we, e.g., denote 10 available machines from A to J, possible process vectors could be (A, D, E), (F, B, F, C), or (D). The minimum length of a process vector is 1, but there is (in theory) no upper limit. Machines can appear in arbitrary order and can appear more than once, if the machine is used multiple times throughout the process.

3.3. Distance Measures for the Quantification of Process Similarity

We define two processes as identical if they have the same length and are composed of the same operations in exactly the same order (e.g., (A, B, A, C) and (A, B, A, C) – but not (A, B, A, C) and (A, B, C, A)). Furthermore, two processes are completely different if the two sets of involved machines are disjoint, no matter how many items they contain (e.g., (A, B) and (C, D, E)). Any combination in between these two extremes is a certain degree of similarity, which needs to be properly

quantified. When selecting applicable distance measures for vectors, three prerequisites have to be considered: (1) the machine IDs in the vectors are nominal values, (2) it must be possible to compare vectors of different lengths, and (3) differences in the order of machines must be less penalized than having completely different machines.

Due to (1) and (2), measures such as the Euclidian distance or Manhattan distance cannot be applied, because these measures require the vectors to be of equal length and the distance between single vector entries can be computed. The Hamming distance for comparison of two strings also requires strings of equal length and thus violates (1). However, in our scenario, e.g., the distance between (A, B) and (A, C) cannot be quantified differently than the distance between (A, B) and (A, F), because machines B and C are not closer than B and F. Prerequisite (3) postulates that (A, B, C) and (A, C, B) must be less different than (A, B, C) and (A, D, E): distance measures that just compare itemwise would assign the same difference, although the first example must be considered as more similar in the context of processes.

The **Jaccard distance** (JD, also known as complementary values Jaccard index or Jaccard similarity coefficient) is the ratio of the size of the intersection of two sets and the size of the union of both sets. To apply JD in our case, we convert the vectors to sets (in which each element occurs only once and the order is not relevant) and determine the ratio of the size intersection and union. E.g., processes (A, A, B, B) and (A, B, C) would be converted to sets [A, B] and [A, B, C] and result in a JD of $2/3 = 0.667$.

The problem of the JD measure is that information about the length of the process is lost during the conversion to a set and also different numbers of occurrences in the different processes are neglected. Therefore, we have developed the **modified Jaccard distance** (MJD), which sums up the number of distinct matching items in two lists and divides by the sum of

items in both lists. E.g., processes (A, A, B, B) and (A, B, C) have four common items (see crossed out items: (~~A~~, A, ~~B~~, B); (~~A~~, ~~B~~, C)) and seven items in total, which yields $4/7 = 0.571$. Like JD, MJD ranges between 0 and 1.

Bose and van der Aalst have identified the Levenshtein distance (LD) as a possible measure to quantify the distance between two processes. It is the minimum number of edit operations (insertion, deletion, or substitution) needed to transform one sequence into the other [18]. We use again a modified version, the **Damerau-Levenshtein distance** (DLD), which also allows a transposition of adjacent items as edit operation [21]. From our perspective, this is an essential modification, because, e.g., processes (A, B) and (B, A) would exhibit a LD of two (one deletion, one insertion), while DLD is only one. In LD, the distance of the pair (A, B) and (B, C) would be identical as in the previous example, while DLD is able to distinguish between the (former) more similar case in contrast to the (latter) more distinct case. LD and DLD can be normalized by multiplication with 2 and division by the summed length of both vectors.

Before we applied the distance measures as part of the clustering procedure, we inspected the behavior of the three measures by counting the frequency of determined distances for each pair of processes in all six datasets. Fig. 1 shows the results. It can be observed that the processes in datasets d and f are very distinct, while b and e exhibit a higher level of similarity. Interestingly, the three measures JD, MJD, and DLD show a high degree of similarity in their distributions. Thus, we conclude that any of the measures is an appropriate candidate for grouping similar process variants together.

3.4. K-Medoids Clustering for the Identification of Process Groups and for the Evaluation of Context Information

The next step in our procedure is to group similar variants of processes together using clustering approaches. Finally, we include context information in the clustering algorithm to improve the results. We have selected k-medoids clustering, which is classified as partitioning clustering, because we can influence the clustering process by selecting the initial medoids to include context information.

Two types of contextual information have been extracted from the data. Firstly, we have counted the frequency of occurrence of each process. E.g., in a scenario with the processes (A, B), (C, D), (A, B), (A, B, C), the frequencies would be (A, B): 2, (C, D): 1, (A, B, C): 1. The underlying assumption here is that the more frequent a process is, the more likely is it to be a standard process, around which process variants exist. Secondly, we determined the stability of each process in terms of its variation in processing time, assuming that stable processes are standard processes. To collect this information, we computed the overall cycle time of each process (by subtracting the start time of the first process operation from the completion time of the last process operation, e.g., $\text{end}(C) - \text{start}(A)$ in case of (A, B, C)). After that, we calculated the coefficient of variation (COV) of the cycle time (i.e. the standard deviation of the cycle time,

normalized by the average cycle time in order to make processes of different duration comparable).

We have conducted three different variants of the k-medoids approach to all six datasets with a range of k from 2 to 30: (1) perform 100 runs of the regular k-medoids with random cluster initialization, (2) run k-medoids using the first k processes with the highest frequency as initial cluster medoids, and (3) run k-medoids using the first k processes with lowest COV as initial cluster medoids. Due to the similarity of the investigated distance measures (see Section 3.3), we have only used DLD for the determination of distance during clustering and skipped the other measures. To evaluate the performance of the three different clustering variants with regard to their ability to identify distinct clusters of process variants, we have used a classical measure for clustering quality in k-means and k-medoids, the summed distance of all items to their respective medoid. The lower this value is, the better is the attribution of items to cluster medoids. Fig. 2 illustrates the results for all six datasets. The error bars of the 'random medoid initialization' curve show the standard deviation over the 100 clustering runs. Missing values indicate that no feasible set of clusters could be found in the corresponding set-up.

As expected, a higher value of k reduces the summed distance, because a higher number of medoids reduces the average distance of every item to the next medoids. This is a regular phenomenon in k-means and k-medoids clustering. Furthermore, we can see in some datasets and for some ranges of k , that the selection of the most frequent processes as medoids outperforms the average values of random medoids selection, in some cases even below the variance corridor. In contrast, a better performance of the cycle time based medoids is extremely rare. Therefore, we conclude that selecting the most frequent processes as main processes and grouping other processes as variants around these processes is a feasible approach to reduce the complexity of the result of Process Mining in heterogeneous process environments in manufacturing and logistics. However, we also have to point out that the results are not fully unambiguous and also the 'process frequency' criterion fails in a variety of cases. An analysis is given in the following Discussion section.

3.5. Discussion

If we look at the results in Fig. 2, we can see that the 'process frequency' method works well in company a for k from 5 to 17, in company b for k from 16 to 21, in company d for k from 14 to 24, in company e only for very large k from 27 on, and in company f for k from 10 to 19. A reason for the variation in the results, especially in the range of k for the sweet spot for the process frequency based approach, lies in the diversity of the datasets. Also, finding the ideal value for k in advance is not possible and it is therefore necessary to check different values of k . With regard to the integration of context information in the Process Mining procedure to manage complex and heterogeneous process environments, the cycle time based stability can be ruled out as relevant context information. Surprisingly, it only performs well to a certain extent in company c, where the process frequency based approach clearly succumbs the random approach.

4. Summary and Conclusion

Rapid changes in logistics and manufacturing cause a high frequency of change and a high number of variants in the industries' processes. A proper documentation of all processes becomes difficult in such a heterogeneous environment and calls for automated documentation approaches. Process Mining offers the possibility to extract process information from event logs, which are usually available in manufacturing and logistics as part of the planning and control infrastructure. This paper aimed at improving the automated mining for process documentation by adding context information to the process detection approach.

In our study, we have investigated how the frequency of occurrence of a process and the stability of the cycle time of repeating processes can be used as additional contextual information when grouping similar processes variants together. We found that selecting the most frequent processes as main processes yields the best results. Additionally, we showed that the three feasible distance measures lead to very similar results when quantifying process similarity, so that no specific effort has to be put in the selection of the appropriate measure.

Future research should focus on the investigation of additional context information that can be retrieved from event logs and on its evaluation in comparison to the frequency of a processes as tested in this paper. Additionally, a methodology to identify the main processes in the first place (i.e. determining the proper value for k) would be beneficial to improve the results of the procedure. Furthermore, additional clustering algorithms that meet the requirements regarding the handling of nominal data should be tested.

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